Question 1 T1

A pizza with a diameter of 15 inches is cut into equal slices, each with a central angle of 36°.
Determine the length of the crust on the outer edge of one slice of pizza.

Solution

$$\frac{36^\circ}{180^\circ} = \frac{\theta}{\pi}$$

$$\theta = \frac{\pi}{5}$$

1 mark for conversion

$$s = \theta r$$

$$s = \left(\frac{\pi}{5}\right)\left(\frac{15}{2}\right)$$

1 mark for substitution

$$s = \frac{3\pi}{2} \text{ inches}$$

or

$$s = 4.712 \text{ inches}$$

2 marks

Question 2 P3

There are 9 girls and 7 boys in a math class from which a committee of 5 is to be chosen.

a) How many different committees of 5 can be formed if one of the boys, William, must be on the committee?

b) How many different committees of 5 can be formed if there must be 2 girls and 3 boys on the committee?

Solution

a) ${}_{1}C_{1} \bullet {}_{15}C_{4} = 1365$

1 mark

b) ${}_{9}C_{2} \bullet {}_{7}C_{3} = 1260$

½ mark for ${}_{9}C_{2}$
½ mark for ${}_{7}C_{3}$
1 mark for the product of combinations

2 marks

Note(s):

- ${}_{1}C_{1}$ does not need to be shown

Question 3 T5

Solve the following equation over the interval $[0, 2\pi]$:

$$\sin^2\theta + 6\sin\theta - 2 = 0$$

Solution

$$\sin\theta = \frac{-6 \pm \sqrt{(6)^2 - 4(1)(-2)}}{2(1)}$$

$$\sin\theta = \frac{-6 \pm \sqrt{36+8}}{2}$$

$$\sin\theta = \frac{-6 \pm \sqrt{44}}{2}$$

$\sin\theta = 0.316\ 624\ldots$	$\sin\theta = -6.316\ 624\ldots$	1 mark for solving for $\sin\theta$
$\theta_r = 0.322\ 169$		
$\theta = 0.322$	no solution	2 marks for solving for θ (½ mark for each value, 1 mark for indicating no solution)
$\theta = 2.819$		

3 marks

Question 4 R10

Solve:

$$6(5)^{3x+2} = 9^{2-x}$$

Solution

$$\log\left[6(5)^{3x+2}\right] = \log 9^{2-x}$$ ½ mark for applying logarithms

$$\log 6 + \log 5^{3x+2} = \log 9^{2-x}$$ 1 mark for product law

$$\log 6 + (3x+2)\log 5 = (2-x)\log 9$$ 1 mark for power law

$$\log 6 + 3x\log 5 + 2\log 5 = 2\log 9 - x\log 9$$

$$3x\log 5 + x\log 9 = 2\log 9 - 2\log 5 - \log 6$$ ½ mark for collecting terms with x

$$x(3\log 5 + \log 9) = 2\log 9 - 2\log 5 - \log 6$$

$$x = \frac{2\log 9 - 2\log 5 - \log 6}{3\log 5 + \log 9}$$ ½ mark for solving for x

$$x = -0.087\ 707$$ ½ mark for evaluating quotient of logarithms

$$= -0.088$$

4 marks

Question 5 T5

Solve $\left(2\sin\theta - 1\right)\left(\sin\theta + 1\right) = 0$ where $\theta \in \mathbb{R}$.

Solution

$2\sin\theta - 1 = 0$	$\sin\theta + 1 = 0$	
$\sin\theta = \dfrac{1}{2}$	$\sin\theta = -1$	1 mark for solving for $\sin\theta$
$\theta_r = \dfrac{\pi}{6}$	$\theta_r = \dfrac{3\pi}{2}$	
$\theta = \dfrac{\pi}{6} + 2k\pi,\ k \in \mathbb{Z}$	$\theta = \dfrac{3\pi}{2} + 2k\pi,\ k \in \mathbb{Z}$	2 marks for solving for θ (1 mark for each branch)
$\theta = \dfrac{5\pi}{6} + 2k\pi,\ k \in \mathbb{Z}$		1 mark for general solution
or	**or**	
$\theta_r = 30^\circ$	$\theta_r = 270^\circ$	
$\theta = 30^\circ + 360^\circ k,\ k \in \mathbb{Z}$	$\theta = 270^\circ + 360^\circ k,\ k \in \mathbb{Z}$	
$\theta = 150^\circ + 360^\circ k,\ k \in \mathbb{Z}$		**4 marks**

Question 6 R12

The roots of the polynomial equation $3(x-2)(x+1)^2 = 0$ are $x = 2$ and $x = -1$.

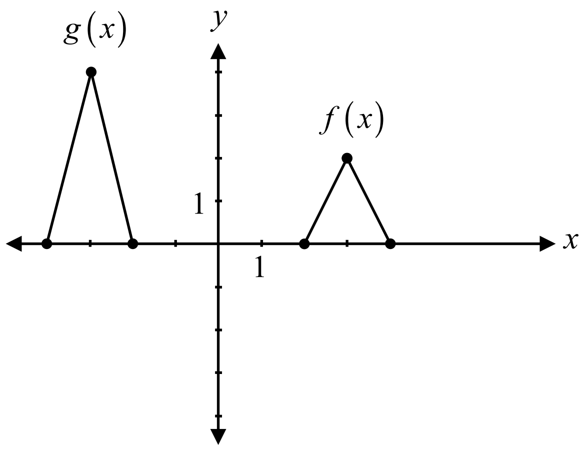
Explain what these roots represent on the graph of $p(x) = 3(x-2)(x+1)^2$.

Solution

They are the x-intercepts of the graph of $p(x)$.

1 mark

Determine an equation for $g(x)$ as a transformation of $f(x)$.



Solution

$g(x) = \underline{2f(x+6)}$

1 mark for vertical stretch
1 mark for horizontal translation

2 marks

or

$g(x) = \underline{2f(-x)}$

1 mark for vertical stretch
1 mark for horizontal reflection

2 marks

Question 8 R11

A student must determine the factors of $5x^4 - 2x^3 + 4x - 1$. He used 5, -2, 4, and -1 as the coefficients of the polynomial when using synthetic division.

Explain the student's error.

Solution

The student did not write the coefficient of 0 for the x^2 term.

1 mark

Question 9 R2, R5

Describe the transformations of $y = f(x)$ when asked to sketch the graph of $y = -f(x - 4)$.

Solution

$f(x)$ is reflected over the x-axis and translated 4 units to the right.

1 mark for vertical reflection
1 mark for horizontal translation

2 marks

Question 10 T6

Prove the identity below for all permissible values of θ:

$$\sin\theta + \frac{\cos\theta}{\tan\theta} = \frac{1}{\cos\theta\tan\theta}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\sin\theta + \frac{\cos\theta}{\tan\theta}$	$\frac{1}{\cos\theta\tan\theta}$
$\sin\theta + \frac{\cos\theta}{\frac{\sin\theta}{\cos\theta}}$	$\frac{1}{\cos\theta\frac{\sin\theta}{\cos\theta}}$
$\sin\theta + \frac{\cos^2\theta}{\sin\theta}$	$\frac{1}{\sin\theta}$
$\frac{\sin^2\theta}{\sin\theta} + \frac{\cos^2\theta}{\sin\theta}$	
$\frac{\sin^2\theta + \cos^2\theta}{\sin\theta}$	
$\frac{1}{\sin\theta}$	

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove an identity

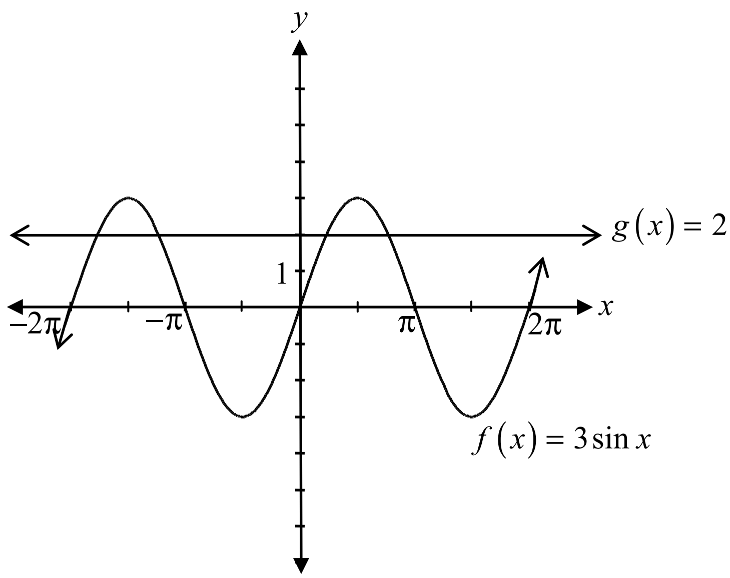
3 marks

Method 2

Left-Hand Side	Right-Hand Side	
$\sin\theta + \dfrac{\cos\theta}{\tan\theta}$	$\dfrac{1}{\cos\theta\tan\theta}$	
	$\dfrac{1}{\cos\theta\dfrac{\sin\theta}{\cos\theta}}$	1 mark for correct substitution of identities
	$\dfrac{1}{\sin\theta}$	
	$\dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta}$	
	$\dfrac{\sin^2\theta}{\sin\theta} + \dfrac{\cos^2\theta}{\sin\theta}$	
	$\sin\theta + \cos\theta\dfrac{\cos\theta}{\sin\theta}$	1 mark for algebraic strategies
	$\sin\theta + \cos\theta\cot\theta$	
	$\sin\theta + \dfrac{\cos\theta}{\tan\theta}$	1 mark for logical process to prove an identity

3 marks

Describe how to use the graphs of $f(x) = 3\sin x$ and $g(x) = 2$ to solve the equation $3\sin x = 2$.



Solution

The solution will be the *x*-values where the two graphs intersect.

1 mark

Question 12 P1

A hockey arena has 5 doors.

Determine the number of ways that you can enter through one door but exit through a different door.

Solution

$\underline{\quad 5 \quad} \bullet \underline{\quad 4 \quad} = 20$ ways

1 mark

Given that $(x+3)$ is one of the factors, express $2x^3+7x^2+2x-3$ as a product of factors.

Solution

$$\begin{array}{r|rrrr} -3 & 2 & 7 & 2 & -3 \\ & \downarrow & -6 & -3 & 3 \\ \hline & 2 & 1 & -1 & 0 \end{array}$$

½ mark for $x=-3$
1 mark for synthetic division (or for any other equivalent strategy)

$(x+3)\left(2x^2+x-1\right)$

or

$(x+3)(2x-1)(x+1)$

½ mark for consistent product of factors

2 marks

Question 14 R12

Identify the maximum number of x-intercepts for a polynomial function of degree 3.

a) 1

b) 2

c) 3

d) 4

Question 15 R3

The graph of $y = f(x)$ contains the point (a, b). The graph of $g(x)$ is a transformation of the graph of $f(x)$ and contains the point $(3a, b)$.

Identify the function that represents $g(x)$.

a) $g(x) = f(3x)$

b) $g(x) = 3f(x)$

c) $g(x) = f\left(\frac{x}{3}\right)$

d) $g(x) = \frac{1}{3}f(x)$

Question 16 T1

The angle 2.95 radians, in standard position, terminates in quadrant:

a) I

b) II

c) III

d) IV

Question 17 T6

Evaluate:

$$2\sin\frac{\pi}{8}\cos\frac{\pi}{8}$$

a) $\frac{1}{2}$

b) $\frac{\sqrt{2}}{2}$

c) 1

d) $\sqrt{2}$

Question 18 P4

Identify which of the following represents the 5th term in the expansion of $\left(4x^2 - 2y^3\right)^{15}$.

a) ${}_{15}C_5\left(4x^2\right)^{10}\left(-2y^3\right)^5$

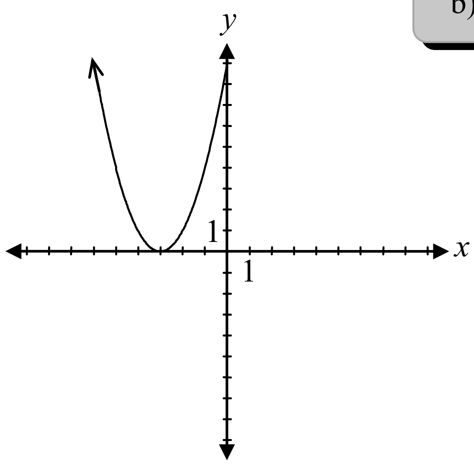
b) ${}_{15}C_5\left(4x^2\right)^{11}\left(-2y^3\right)^4$

c) ${}_{15}C_4\left(4x^2\right)^{10}\left(-2y^3\right)^5$

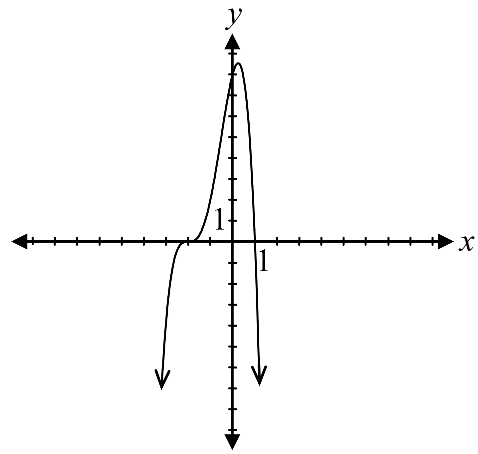
d) ${}_{15}C_4\left(4x^2\right)^{11}\left(-2y^3\right)^4$

Identify which of the following graphs of polynomial functions has a zero with a multiplicity of 3.

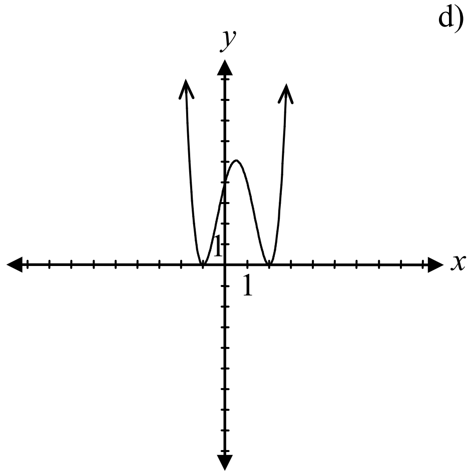
a)



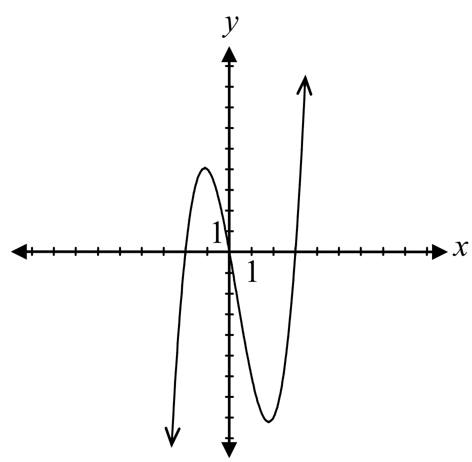
b)



c)



d)



Question 20 T6

A non-permissible value of x for the function $f(x) = \dfrac{1}{\cos x + 1}$ is:

a) -1

b) 0

c) π

d) $\dfrac{3\pi}{2}$

Question 21 R14

Identify which of the following statements is true for the rational function $f(x) = \dfrac{4(x-1)(x-2)}{(x-1)(x+3)}$.

a) The equation of the horizontal asymptote is $y = 4$.

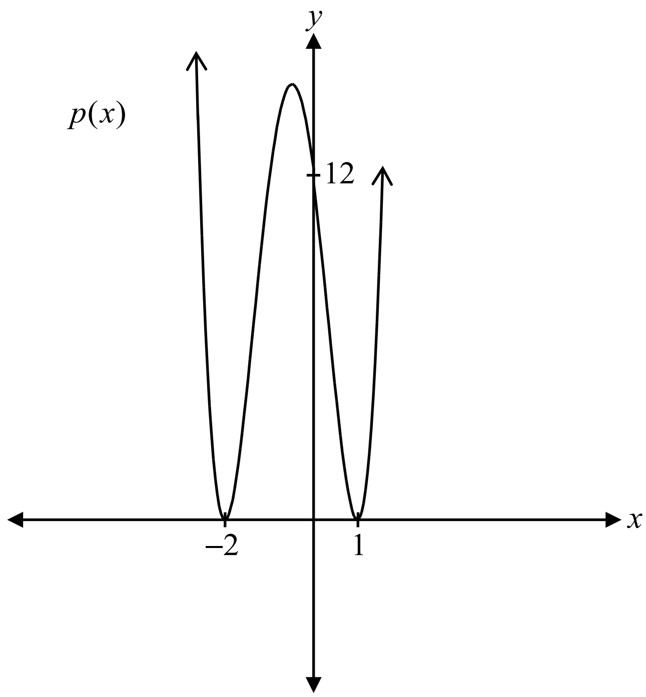
b) The equation of the vertical asymptote is $x = 1$.

c) The y-intercept is 0.

d) There is a point of discontinuity (hole) when $x = 2$.

Question 22 R12

Determine the equation of the polynomial function, $p(x)$, represented by the graph.



Solution

$p(x) = \underline{3(x+2)^2(x-1)^2}$

1 mark for factors

1 mark for multiplicity of 2 (½ mark for each)

1 mark for correct value of a (consistent with factors and multiplicity)

3 marks

Question 23 R7

Evaluate:

$$\log_4 2$$

Solution

$\frac{1}{2}$ **1 mark**

Question 24 T3

Evaluate:

$$\left(\cos\frac{11\pi}{3}\right)\left(\csc\frac{11\pi}{6}\right)$$

Solution

$$\left(\frac{1}{2}\right)(-2)$$

1 mark for $\cos\frac{11\pi}{3}$ (½ mark for the quadrant, ½ mark for the value)

1 mark for $\csc\frac{11\pi}{6}$ (½ mark for the quadrant, ½ mark for the value)

$$-1$$

2 marks

Question 25 R7

Estimate the value of $\log_2 5$.

Justify your answer.

Solution

$$\left.\begin{aligned}\log_2 4 &= 2\\ \log_2 8 &= 3\end{aligned}\right\}$$ ½ mark for justification

$\therefore \log_2 5 \approx 2.3$ ½ mark for estimated value in the interval $[2.1, 2.4]$

1 mark

Question 26 T2

If θ terminates in quadrant III and $\cos\theta = -\frac{6}{7}$, determine the exact value of $\tan\theta$.

Solution

$$\cos\theta = \frac{x}{r}$$

$$x^2 + y^2 = r^2$$

$$y^2 = (7)^2 - (-6)^2$$ ½ mark for substitution of $x = -6$ and $r = 7$

$$y^2 = 13$$

$$y = \pm\sqrt{13}$$ ½ mark for solving for y

$$\tan\theta = \frac{\sqrt{13}}{6}$$ 1 mark for the value of $\tan\theta$ (½ mark for the quadrant, ½ mark for the value)

2 marks

Given $f(x) = x^2 + x - 4$ and $g(x) = \sqrt{x+5}$, Taz was asked to find $f(g(x))$.

Taz's solution:

$$f(g(x)) = \left(\sqrt{x+5}\right)^2 + x - 4$$
$$= x + 5 + x - 4$$
$$= 2x + 1, \ x \geq -5$$

Describe the error in Taz's solution.

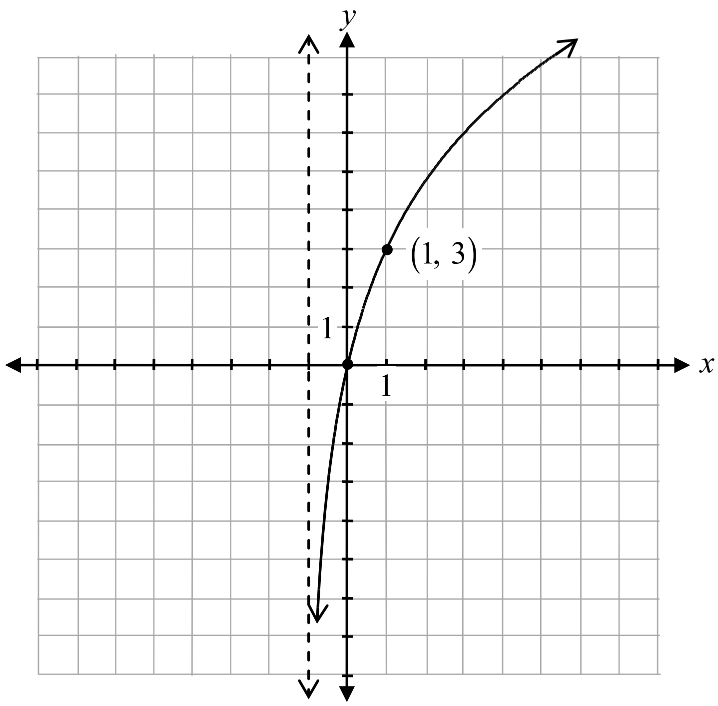
Solution

Taz must substitute $g(x)$ in both terms containing x in $f(x)$ and then simplify.

1 mark

Sketch the graph of the function $f(x) = 3\log_2(x+1)$.

Solution



1 mark for increasing logarithmic function
1 mark for vertical stretch
1 mark for asymptotic behaviour at $x = -1$

3 marks

Question 29 R14

Write an equation of a rational function that would not have any vertical asymptotes.

Solution

Various equations, such as the following, are possible:

$$y = \frac{(x-2)(x+1)}{(x-2)}$$

or

$$y = \frac{4}{x^2+4}$$

1 mark

Question 30 T6

Determine the exact value of $\tan 75^\circ$.

Solution

$\tan 75^\circ = \tan\left(30^\circ + 45^\circ\right)$

1 mark for combination

$$= \frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \tan 45^\circ}$$

$$= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}(1)}$$

1 mark for exact values (½ mark for each)

2 marks

$$= \frac{\frac{1+\sqrt{3}}{\sqrt{3}}}{\frac{\sqrt{3}-1}{\sqrt{3}}}$$

$$= \frac{1+\sqrt{3}}{\sqrt{3}-1}$$

or

$$= \frac{\sqrt{3}+3}{3-\sqrt{3}}$$

Note(s):

- Other combinations are possible.

Sketch the graph of the following function:

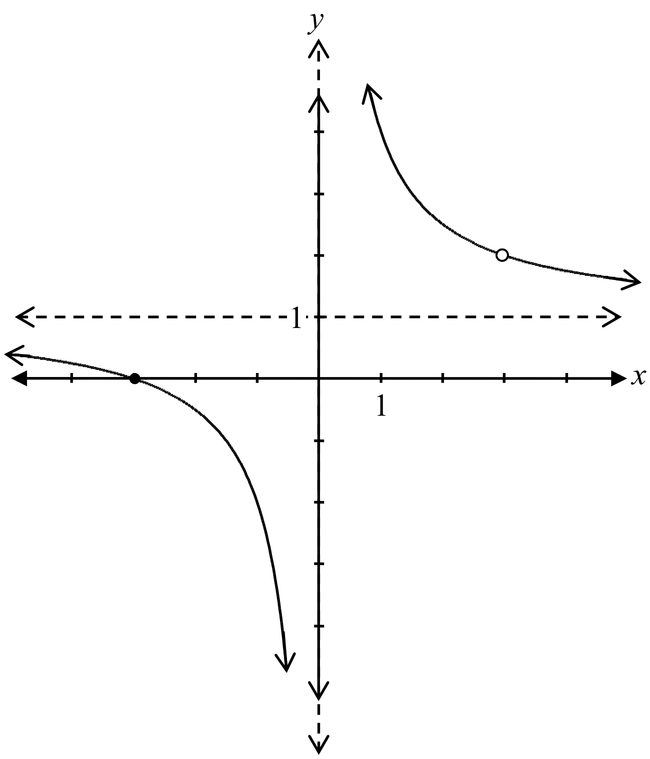
$$f(x) = \frac{(x+3)(x-3)}{x(x-3)}$$

Solution

$$f(x) = \frac{(x+3)\cancel{(x-3)}}{x\cancel{(x-3)}}$$

$$= \frac{x+3}{x}, \; x \neq 3$$

$\therefore$ there is a point of discontinuity (hole) at $(3,\ 2)$



1 mark for asymptotic behaviour at $y = 1$
1 mark for asymptotic behaviour at $x = 0$
1 mark for point of discontinuity (hole) at $(3,\ 2)$
(½ mark for $x = 3$, ½ mark for $y = 2$)
½ mark for graph left of vertical asymptote
½ mark for graph right of vertical asymptote

4 marks

In the binomial expansion of $\left(\frac{1}{x^3} - 2x^2\right)^9$, determine which term contains x^3.

Solution

Method 1

$x^3 = \left(x^{-3}\right)^{9-k}\left(x^2\right)^k$

½ mark for substitution

$x^3 = x^{-27+3k+2k}$

$3 = -27 + 3k + 2k$

$30 = 5k$

$6 = k$

½ mark for solving for k

$\therefore$ term 7 would contain x^3

1 mark for identifying the 7th term
(or consistent term with the value of k)

2 marks

Method 2

$\left(\frac{1}{x^3}\right)^9, \left(\frac{1}{x^3}\right)^8\left(x^2\right), \left(\frac{1}{x^3}\right)^7\left(x^2\right)^2$

$x^{-27}, x^{-22}, x^{-17}$

1 mark for determining the pattern

$\therefore$ term 7 would contain x^3

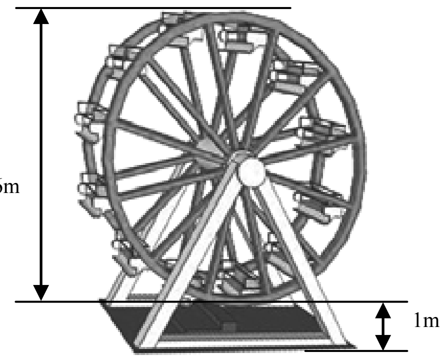
1 mark for identifying the 7th term
(or consistent term with the pattern)

2 marks

Question 33 T4

José and Dana get on a Ferris wheel, which is 1 metre off the ground. The diameter of the Ferris wheel is 16 metres. Their ride lasts for 4 minutes, in which time the Ferris wheel makes one revolution.

Determine the values of A, B, C, and D, if the sinusoidal function that models the situation is $h(t) = A\cos[B(t-C)] + D$, where h is the height at which José and Dana are located on the Ferris wheel, from the ground, in metres, and t is the time, in minutes.



A = ________________

B = ________________

C = ________________

D = ________________

Solution

$A = 8$	or	$A = -8$	1 mark for A
$B = \frac{\pi}{2}$		$B = \frac{\pi}{2}$	1 mark for B
$C = 2$		$C = 0$	1 mark for C
$D = 9$		$D = 9$	1 mark for D

4 marks

Note(s)

- Other answers are possible.

Question 34 P2

Solve algebraically:

$$_{n}P_{3} = 4!(n-1)$$

Solution

$$\frac{n!}{(n-3)!} = 4!(n-1)$$

½ mark for substitution

$$\frac{n\cancel{(n-1)}(n-2)\cancel{(n-3)}!}{\cancel{(n-3)}!} = 4!\cancel{(n-1)}$$

1 mark for factorial expansion
½ mark for simplification of factorials

$$n(n-2) = 24$$

$$n^2 - 2n - 24 = 0$$

$$(n-6)(n+4) = 0$$

$$n = 6 \quad \xcancel{n = -4}$$

½ mark for rejecting extraneous root
½ mark for the value of *n*

3 marks

Given $f(x) = \dfrac{2}{x-1}$, determine the equation of the inverse, $f^{-1}(x)$.

Solution

Method 1

let $y = f(x)$

$$f(x) = \frac{2}{x-1}$$

$$y = \frac{2}{x-1}$$

$$x = \frac{2}{y-1}$$

1 mark for switching x and y values

$$y - 1 = \frac{2}{x}$$

$$y = \frac{2}{x} + 1$$

½ mark for solving for y

$$f^{-1}(x) = \frac{2}{x} + 1$$

½ mark for writing equation of $f^{-1}(x)$

2 marks

Method 2

let $y = f(x)$

$$f(x) = \frac{2}{x-1}$$

$$y = \frac{2}{x-1}$$

$$x = \frac{2}{y-1}$$

1 mark for switching x and y values

$$x(y-1) = 2$$

$$xy - x = 2$$

$$xy = 2 + x$$

$$y = \frac{2+x}{x}$$

½ mark for solving for y

$$f^{-1}(x) = \frac{2+x}{x}$$

½ mark for writing equation of $f^{-1}(x)$

2 marks

Solve:

$$4\log_3 2 - \frac{1}{3}\log_3 8 = \log_3 a$$

Solution

Method 1

$\log_3 2^4 - \log_3 8^{\frac{1}{3}} = \log_3 a$	1 mark for power law (½ mark for each)
$\log_3 16 - \log_3 2 = \log_3 a$	
$\log_3\left(\frac{16}{2}\right) = \log_3 a$	1 mark for quotient law
$\log_3 8 = \log_3 a$	
$a = 8$	1 mark for equating arguments

3 marks

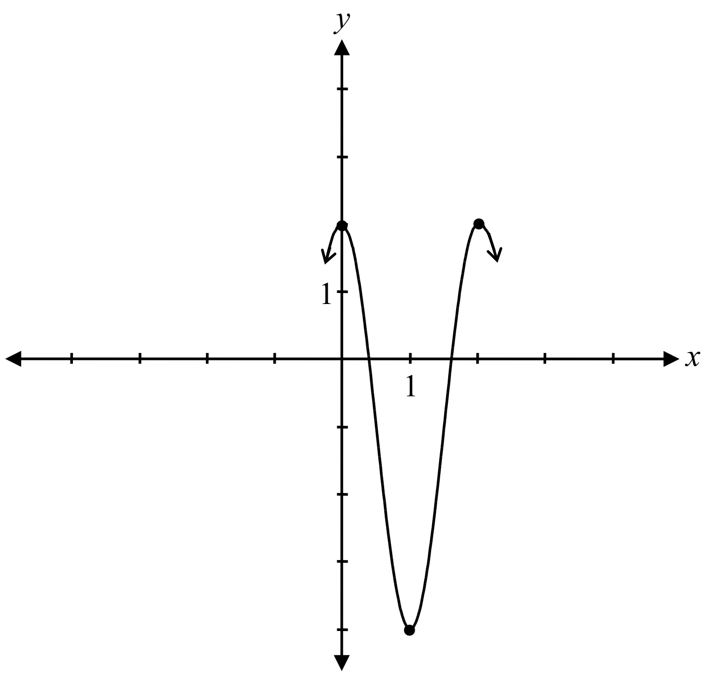
Method 2

$\log_3 2^4 - \log_3 8^{\frac{1}{3}} - \log_3 a = 0$	1 mark for power law (½ mark for each)
$\log_3 16 - \log_3 2 - \log_3 a = 0$	
$\log_3\left(\frac{16}{2a}\right) = 0$	1 mark for quotient law
$\log_3\left(\frac{8}{a}\right) = 0$	
$3^0 = \frac{8}{a}$	1 mark for exponential form
$1 = \frac{8}{a}$	
$a = 8$	

3 marks

Sketch the graph of at least one period of the function $y = 3\cos(\pi x) - 1$.

Solution



$$\text{period} = \frac{2\pi}{\pi}$$
$$= 2$$

1 mark for amplitude
1 mark for period
1 mark for vertical translation

3 marks

Question 38 R8

Using the laws of logarithms, fully expand the expression:

$$\log_a\left(\frac{x^3}{y\sqrt{z}}\right)$$

Solution

$$\log_a\left(\frac{x^3}{y\sqrt{z}}\right) = \log_a x^3 - \left(\log_a y + \log_a \sqrt{z}\right)$$

1 mark for quotient law
1 mark for product law

$$= 3\log_a x - \left(\log_a y + \frac{1}{2}\log_a z\right)$$

1 mark for power law (½ mark for each)

$$= 3\log_a x - \log_a y - \frac{1}{2}\log_a z$$

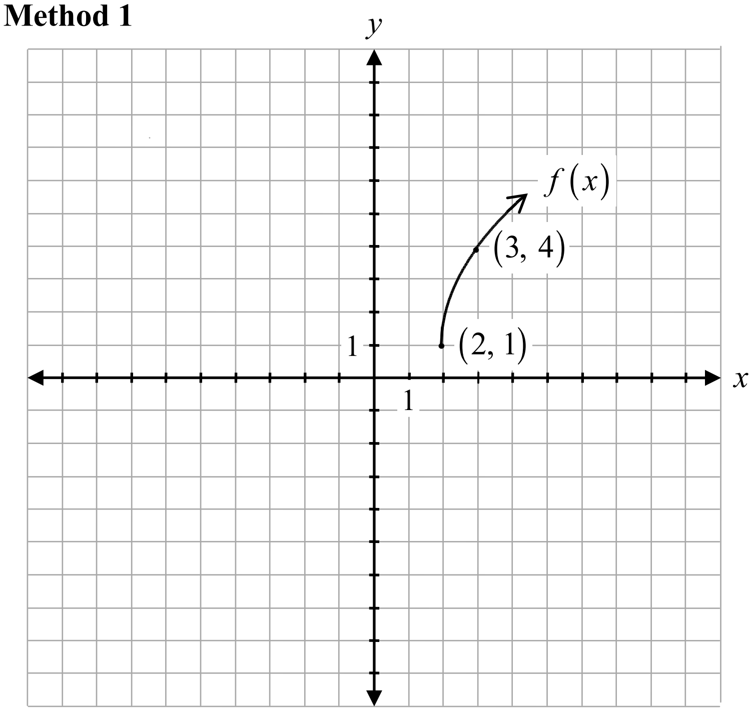
3 marks

Question 39 R13

Sketch the graph of $f(x) = 3\sqrt{x-2} + 1$.

Solution

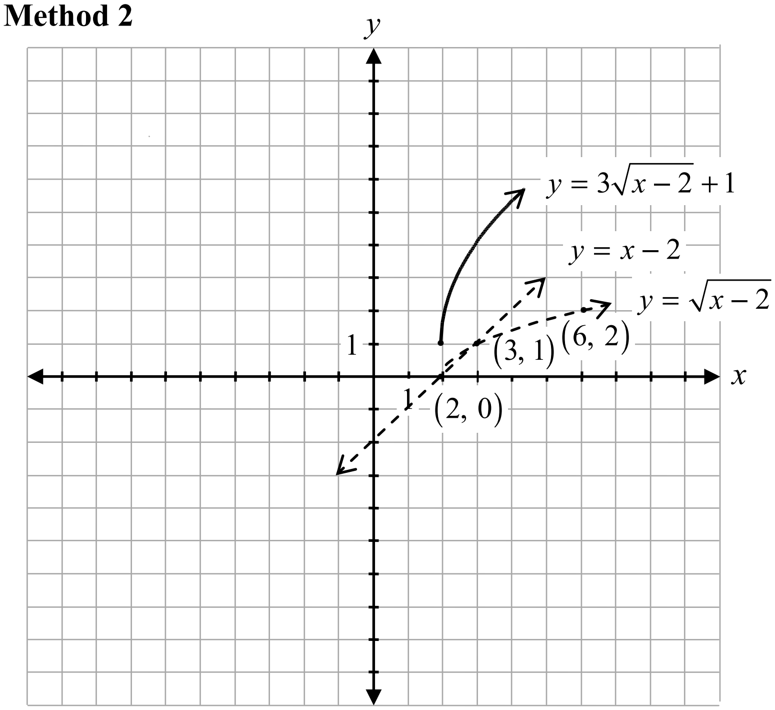
Method 1



1 mark for horizontal translation
1 mark for vertical translation
1 mark for shape of a radical function
1 mark for vertical stretch

4 marks

Method 2



1 mark for invariant points where $y = 0$ and $y = 1$ (½ mark for each point)
1 mark for domain $[2, \infty)$
½ mark for shape between invariant points
½ mark for shape to the right of the invariant points
1 mark for applying transformations (½ mark for vertical stretch, ½ mark for vertical translation)

4 marks

Question 40 R13

a) Determine the domain of the graph of the function $f(x) = \sqrt{x^2 - 4}$.

b) Explain why the domain of $f(x) = \sqrt{x^2 - 4}$ is restricted.

Solution

a) $\{x \mid x \le -2 \quad \cup \quad x \ge 2\}$

or

D: $(-\infty, -2] \cup [2, \infty)$

1 mark for domain (½ mark for $x \le -2$, ½ mark for $x \ge 2$)

1 mark

b) The domain is restricted because you cannot take the square root of a negative number.

1 mark

Question 41 R1, R2, R5

Given the point $(-12, -18)$ on the graph of $f(x)$, determine the new points after the following transformations of $f(x)$.

a) $\dfrac{1}{f(x)}$

b) $f(-x)+10$

Solution

a) $\left(-12, \dfrac{-1}{18}\right)$

1 mark

b) $(12, -8)$

1 mark (½ mark for x-value, ½ mark for y-value)

1 mark

Explain why there is no solution for the equation $\csc\theta = -\frac{1}{2}$.

Solution

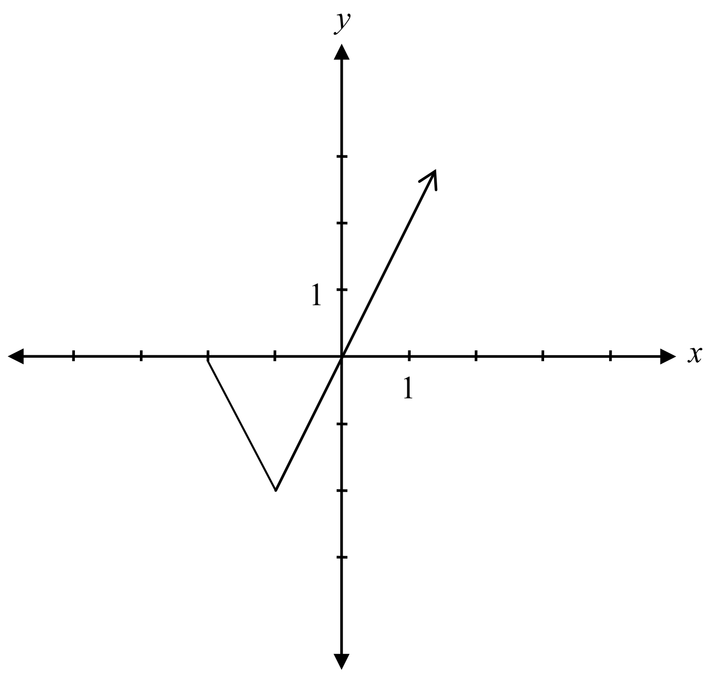
The value of $\csc\theta$ cannot be between -1 and 1.

or

The value of $\sin\theta$ cannot be less than -1.

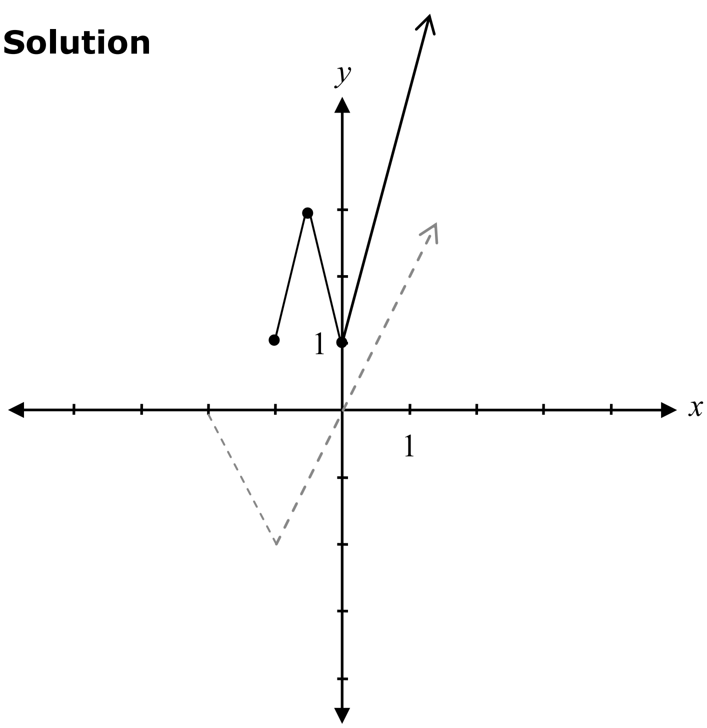
1 mark

Given the graph of $y = f(x)$,



sketch the graph of $y = \left|f(2x)\right| + 1$.

Solution



1 mark for absolute value
1 mark for horizontal compression
1 mark for vertical translation

3 marks

Question 44 R9

Given $f(x) = 2^x + 1$, state the equation of the horizontal asymptote.

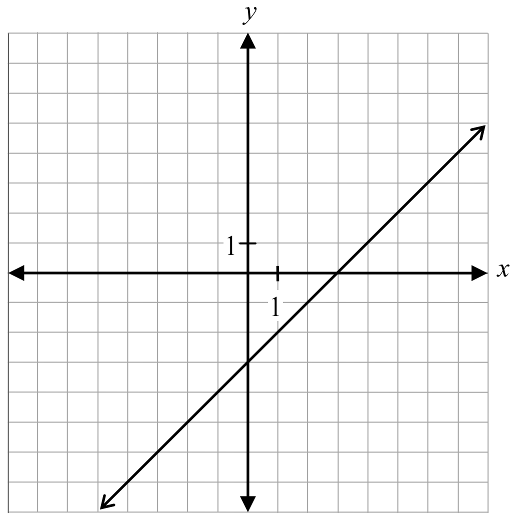
Solution

$y = 1$

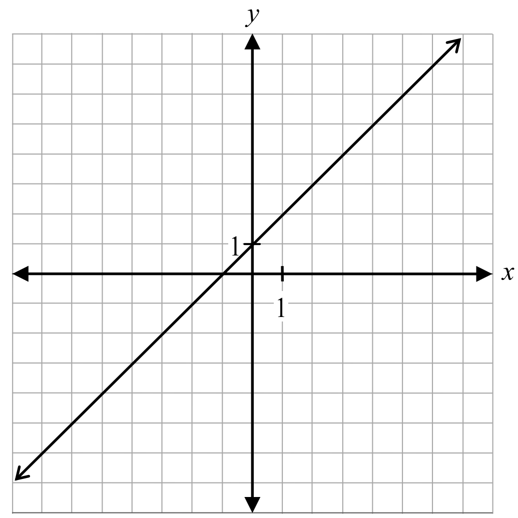
1 mark

Given the following graphs of $f(x) = x - 3$ and $g(x) = x + 1$,

$f(x)$

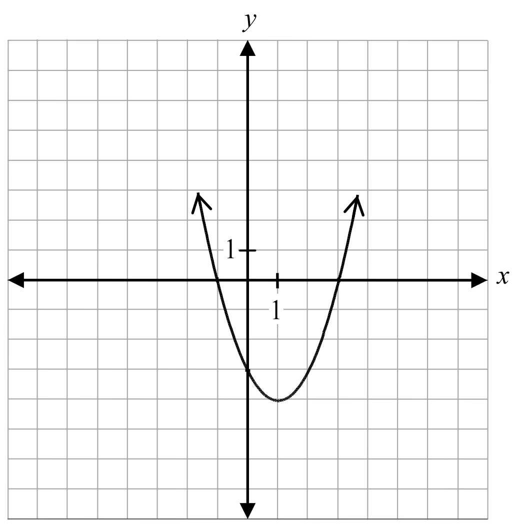


$g(x)$



sketch the graph of $h(x) = (f \bullet g)(x)$.

Solution



1 mark for operation of multiplication
1 mark for shape representing the given operation

2 marks